

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Computer Vision Engineer

Computer vision engineer with experience building and shipping real-time C++ perception systems across edge and cloud-based microservice deployments. Delivered low-latency video analytics pipelines for industrial clients, with production hardening through Docker, Kafka, and observability that cut crash frequency in live systems by 67%. Comfortable optimising CUDA and TensorRT inference under tight latency and memory constraints, and configuring SLAM and pose-graph optimisation tooling (Ceres, RTAB-Map) for sensor fusion and mapping work.

PROJECTS

DeepStream RTSP Pipeline

Jun 2026

- Built an end-to-end **NVIDIA DeepStream 9.0/GStreamer RTSP video analytics pipeline** for multi-stream computer vision, using rtspsrc, nvstreammux, nvinfer, nvtracker, and nvdsosd to run YOLO+NvDCF tracking, metadata extraction, anonymisation, CSV logging, and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a **YOLO26n TensorRT FP16 detector** on **6GB VRAM Turing hardware**, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling, and FP16-vs-FP32 accuracy validation to sustain **29.7 FPS per stream live** and **131.3 FPS per stream unthrottled**.
- Implemented a **C++/CUDA TensorRT decode plugin** using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked tracker configurations across **IOU, NvDCF, and** ByteTrack-inspired **NvSORT**, producing MOTA, IDF1, ID switches, fragmentation, per-frame detection counts, FPS, and VRAM impact to guide production tracker selection.
- Added production hardening through **GPU smoke tests, model-promotion gates, structured JSON logging, and per-camera health metrics** for reactive debugging across liveness, dropped frames, reconnect events, and RSS.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on Bristish Petroleum rigs.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Led on-site validation and diagnostics on deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.
- Presented performance findings to leadership and multinational stakeholders, from project managers to drill rig operators, translating technical results into deployment decisions.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a team to design, train, and deploy vision pipelines for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries.
- Built low-latency asynchronous perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel hardware.

- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Deployed quantised vision models using FP16/INT8 optimisation, pruning and layer fusion, reducing model size by 50% while preserving production accuracy targets.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Configured Ceres, g2o, and GTSAM as solver backends and tuned ICP parameters across environments.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, with tf2-based frame management and message synchronisation to resolve transform timing issues.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **SLAM & 3D Geometry:** RTAB-Map, Cartographer, LIO-SAM, LiDAR SLAM, Scan matching, Pose graph optimisation, Ceres, g2o, GTSAM, ICP (Iterative Closest Point), LiDAR extrinsics calibration
- **Edge AI & Video Pipelines:** NVIDIA DeepStream 6.x/9.0, GStreamer, RTSP video pipelines, CUDA, JetPack, Jetson AGX Xavier, multi-stream video analytics, camera behaviour debugging, ISP tuning, FFmpeg
- **Edge Inference and Optimisation:** OpenVINO, ONNX, PyTorch, TensorFlow, Quantisation-Aware Training, Post-training Quantisation, Structured pruning, Layer fusion, FP16/INT8 deployment, custom TensorRT plugins, latency/throughput profiling, TensorRT
- **Computer Vision Models:** Object detection, object tracking, YOLO (YOLOv8, YOLO26), RT-DETR, SSD MobileNet V2, EfficientDet, ByteTrack, NvDCF, segmentation, image processing
- **Cloud, Edge & MLOps:** Microsoft Azure, Apache Kafka, MLflow, DVC (Data Version Control), Docker, CI/CD, Production monitoring, Structured logging, Health checks, model-promotion gates, failure-mode analysis, observability
- **Programming & Tooling:** Python, C++17, Bash, CMake, Git, Linux, Intel VTune, OpenCV, Point Cloud Library (PCL)
- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux